NEET UG CHEMISTRY STANDARD

Hydrocarbons

Syllabus Context: Organic Chemistry — Classification, Nomenclature, Isomerism, Preparation, Physical & Chemical Properties of Alkanes, Alkenes, Alkynes, and Aromatic Hydrocarbons.

1. Learning Objectives & Chapter Overview

After studying this tutorial, you will be able to:

- Name hydrocarbons systematically according to the IUPAC system of nomenclature and recognise structural, chain, position, geometrical, and conformational isomers.
- Understand laboratory and industrial methods of preparation of alkanes, alkenes, alkynes, and benzene.
- Distinguish between alkanes, alkenes, alkynes, and aromatic hydrocarbons on the basis of physical and chemical properties.
- Draw and differentiate between eclipsed, staggered, and skew conformations of ethane using Sawhorse and Newman projections.
- Predict addition products of unsymmetrical alkenes and alkynes using Markovnikov's rule and the anti-Markovnikov peroxide (Kharash) effect based on electronic mechanisms.
- Comprehend the resonance structure of benzene, apply Hückel's $(4n+2)\,\pi$ rule for aromaticity, and explain the mechanism of electrophilic aromatic substitution (S_E).
- Predict the directive influence and activating/deactivating nature of substituents in monosubstituted benzene rings, and understand the carcinogenicity of polynuclear hydrocarbons.

Industrial & Everyday Importance of Hydrocarbons

The term **hydrocarbon** means compounds composed of carbon and hydrogen only. They are the principal sources of energy and industrial raw materials:

- **LPG (Liquefied Petroleum Gas):** Domestic fuel causing the least pollution.
- **CNG (Compressed Natural Gas) & LNG (Liquefied Natural Gas):** Natural gas found in upper strata during drilling of oil wells is compressed to form CNG or liquefied to form LNG.
- **Fractional Distillation of Petroleum:** Yields petrol, diesel, and kerosene oil. Petrol and CNG vehicles cause less pollution. Lower petroleum fractions are used for dry cleaning clothes to remove non-polar grease stains.
- **Destructive Distillation of Coal:** Yields coal gas and coal tar (commercial source of benzene).
- **Polymers & Chemicals:** Starting materials for polymers (polythene, polypropene, polystyrene), solvents for paints, and synthesis of dyes and drugs.

2. Classification of Hydrocarbons

Depending upon the types of carbon-carbon bonds present, hydrocarbons are classified into three main categories:

1. **Saturated Hydrocarbons:** Contain only carbon-carbon and carbon-hydrogen single (σ) bonds.
   - *Alkanes (Paraffins):* Open-chain saturated hydrocarbons with general formula $\mathrm{C_nH_{2n+2}}$.
   - *Cycloalkanes (Alicyclic):* Carbon atoms form a closed chain or ring.
2. **Unsaturated Hydrocarbons:** Contain carbon-carbon multiple bonds — double bonds (alkenes, $\mathrm{C_nH_{2n}}$), triple bonds (alkynes, $\mathrm{C_nH_{2n-2}}$), or both.
3. **Aromatic Hydrocarbons (Arenes):** Special cyclic compounds containing delocalised π-electron clouds (benzenoids and non-benzenoids).

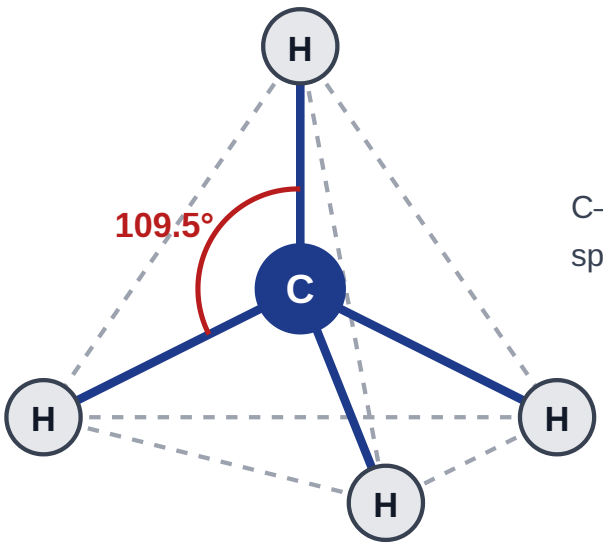
3. Alkanes (Saturated Open-Chain Hydrocarbons)

3.1 Structure, Bonding, and Homologous Series

Alkanes are inert under normal conditions towards acids, bases, and other reagents; hence they were earlier called **paraffins** (Latin: *parum* = little, *affinis* = affinity). Their general formula is $\mathrm{C_nH_{2n+2}}$.

- **Methane ($\mathrm{CH_4}$):** Found in coal mines and marshy places. According to VSEPR theory, methane has a regular **tetrahedral structure** with carbon at the centre and four hydrogens at the four corners. All $\mathrm{H-C-H}$ bond angles are 109.5°.
- **Orbital Overlap & Bond Lengths:** $\mathrm{C-C}$ and $\mathrm{C-H}$ σ bonds are formed by head-on overlapping of sp^3 hybrid orbitals of carbon and $1s$ orbitals of hydrogen. In alkanes, the $\mathrm{C-C}$ bond length is $154\,\mathrm{pm}$ and the $\mathrm{C-H}$ bond length is $112\,\mathrm{pm}$.
- **Classification of Carbon Atoms:**
  - **Primary (1°):** Attached to no other carbon (as in $\mathrm{CH_4}$) or to only 1 carbon atom (terminal carbons are always 1°).
  - **Secondary (2°):** Attached to 2 carbon atoms.
  - **Tertiary (3°):** Attached to 3 carbon atoms.
  - **Quaternary (4° or *neo*):** Attached to 4 carbon atoms.
- **Alkyl Groups:** Derived by removing one hydrogen atom from an alkane; general formula $\mathrm{C_nH_{2n+1}}$ (e.g., $-\mathrm{CH_3}$ methyl, $-\mathrm{C_2H_5}$ ethyl, $-\mathrm{C_3H_7}$ propyl).

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Tutorial Diagram 1: Tetrahedral structure of methane showing 109.5° bond angles.

3.2 Nomenclature & Chain Isomerism in Alkanes

Methane, ethane, and propane have only one structure each. Higher alkanes exhibit **structural (chain) isomerism** due to differences in the carbon chain skeleton:

- **Butane (C_4H_{10}) has 2 chain isomers:**
  1. $CH_3-CH_2-CH_2-CH_3$: Butane (*n*-butane), b.p. $273\,K$ (or $272.4\,K$).
  2. $CH_3-CH(CH_3)-CH_3$: 2-Methylpropane (isobutane), b.p. $261\,K$.
- **Pentane (C_5H_{12}) has 3 chain isomers:**
  1. $CH_3-CH_2-CH_2-CH_2-CH_3$: Pentane (*n*-pentane), b.p. $309\,K$ ($309.1\,K$).
  2. $CH_3-CH(CH_3)-CH_2-CH_3$: 2-Methylbutane (isopentane), b.p. $301\,K$ ($300.9\,K$).
  3. $CH_3-C(CH_3)_2-CH_3$: 2,2-Dimethylpropane (neopentane), b.p. $282.5\,K$.
- **Number of Chain Isomers in Higher Alkanes:** C_6H_{14} has **5** isomers, C_7H_{16} has **9** isomers, and $C_{10}H_{22}$ has **75** possible isomers.

Important IUPAC Nomenclature Examples & Rules

Structure	IUPAC Name	Key Nomenclature Rule Applied
$^{1}CH_3-^{2}CH(CH_3)-^{3}CH_2-^{4}CH(C_2H_5)-^{5}CH_2-^{6}CH_3$	4-Ethyl-2-methylhexane	Lowest locant sum (2,4 vs 3,5) and alphabetical listing of substituents.
$^{8}CH_3-^{7}CH_2-^{6}CH_2-^{5}CH(CH(CH_3)_2)-^{4}CH(CH_3)-^{3}C(C_2H_5)_2-^{2}CH_2-^{1}CH_3$	3,3-Diethyl-5-isopropyl-4-methyloctane	Lowest locant set and alphabetical order (diethyl → e, isopropyl → i, methyl → m).
$^{1}CH_3-^{2}CH_2-^{3}CH_2-^{4}CH(CH(CH_3)_2)-^{5}CH(CH(CH_3)CH_2CH_3)-(CH_2)_4-^{10}CH_3$	5-*sec*-Butyl-4-isopropyldecane	Hyphenated prefix *sec-* is ignored in alphabetisation ('b' of butyl vs 'i' of isopropyl); *isopropyl* is treated as one word.
$^{1}CH_3-^{2}CH_2-^{3}CH_2-^{4}CH_2-^{5}CH(CH_2C(CH_3)_3)-^{6}CH_2-^{7}CH_2-^{8}CH_2-^{9}CH_3$	5-(2,2-Dimethylpropyl)nonane	Complex side chain is numbered from the carbon attached to the main chain as C^1.
$^{1}CH_3-^{2}CH_2-^{3}CH(C_2H_5)-^{4}CH_2-^{5}CH(CH_3)-^{6}CH_2-^{7}CH_3$	3-Ethyl-5-methylheptane	When both ends give the same locants (3,5), alphabetical priority gets the lower number (ethyl at 3, methyl at 5).

3.3 Isomeric Pentyl Groups ($-C_5H_{11}$) and Corresponding Alcohols

Attachment of an $-OH$ group to the 8 different isomeric $-C_5H_{11}$ alkyl structures yields 8 isomeric alcohols:

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Structure of $-C_5H_{11}$ Group	Corresponding Alcohol	IUPAC Name of Alcohol
$CH_3-CH_2-CH_2-CH_2-CH_2-$	$CH_3-CH_2-CH_2-CH_2-CH_2-OH$	Pentan-1-ol
$CH_3-CH(\vert)-CH_2-CH_2-CH_3$	$CH_3-CH(OH)-CH_2-CH_2-CH_3$	Pentan-2-ol
$CH_3-CH_2-CH(\vert)-CH_2-CH_3$	$CH_3-CH_2-CH(OH)-CH_2-CH_3$	Pentan-3-ol
$CH_3-CH(CH_3)-CH_2-CH_2-$	$CH_3-CH(CH_3)-CH_2-CH_2-OH$	3-Methylbutan-1-ol
$CH_3-CH_2-CH(CH_3)-CH_2-$	$CH_3-CH_2-CH(CH_3)-CH_2-OH$	2-Methylbutan-1-ol
$CH_3-C(CH_3)(\vert)-CH_2-CH_3$	$CH_3-C(CH_3)(OH)-CH_2-CH_3$	2-Methylbutan-2-ol
$CH_3-C(CH_3)_2-CH_2-$	$CH_3-C(CH_3)_2-CH_2OH$	2,2-Dimethylpropan-1-ol
$CH_3-CH(CH_3)-CH(\vert)-CH_3$	$CH_3-CH(CH_3)-CH(OH)-CH_3$	3-Methylbutan-2-ol

3.4 Methods of Preparation of Alkanes

1. **Hydrogenation of Unsaturated Hydrocarbons:** Dihydrogen gas adds to alkenes and alkynes in the presence of finely divided catalysts like Pt, Pd, or Ni. Pt and Pd catalyse the reaction at room temperature, whereas Ni requires relatively higher temperature and pressure (Sabatier-Senderens reaction).

$$CH_2=CH_2+H_2 \xrightarrow{Pt/Pd/Ni} CH_3-CH_3$$
$$CH_3-C\equiv C-H+2H_2 \xrightarrow{Pt/Pd/Ni} CH_3-CH_2-CH_3$$

2. **From Alkyl Halides:**
   - *Reduction with Zinc and Dilute Hydrochloric Acid:* Alkyl halides (except fluorides, as $C-F$ bond is very strong) undergo reduction to alkanes:

$$R-Cl+H_2 \xrightarrow{Zn,H^+} R-H+HCl$$

   - *Wurtz Reaction:* Alkyl halides on treatment with metallic sodium in **dry ether** (free from moisture, as sodium reacts violently with water) give higher alkanes containing an **even number of carbon atoms**:

$$2R-Br+2Na \xrightarrow{\text{dry ether}} R-R+2NaBr$$

   Limitation: If two different alkyl halides ($R-X$ and $R'-X$) are used to prepare an alkane with an odd number of carbon atoms, a mixture of three alkanes ($R-R, R'-R'$, and $R-R'$) is formed, which have close boiling points and are difficult to separate. Methane cannot be prepared by Wurtz reaction.

3. **From Carboxylic Acids:**
   - *Decarboxylation with Soda Lime:* Heating sodium salts of carboxylic acids with **soda lime** (a mixture of $NaOH$ and CaO) yields an alkane with **one carbon atom less** than the parent acid:

$$R-COO^-Na^+ + NaOH \xrightarrow[\Delta]{CaO} R-H+Na_2CO_3$$

   For example, preparing propane (C_3H_8) requires the sodium salt of **butanoic acid** ($CH_3CH_2CH_2COO^-Na^+$).
   - *Kolbe's Electrolytic Method:* Electrolysis of a concentrated aqueous solution of sodium or potassium salt of a carboxylic acid gives an alkane containing an **even number of carbon atoms** at the **anode**:

$$2CH_3COO^-Na^+ + 2H_2O \xrightarrow{\text{Electrolysis}} CH_3-CH_3\,(\text{anode}) + 2CO_2\,(\text{anode}) + H_2\,(\text{cathode}) + 2NaOH$$

   Mechanism:
   - At Anode (Oxidation via free radicals): $2CH_3COO^- \xrightarrow{-2e^-} 2CH_3COO^\bullet \longrightarrow 2^\bullet CH_3 + 2CO_2\uparrow$; then $^\bullet CH_3 + ^\bullet CH_3 \longrightarrow CH_3-CH_3\uparrow$.
   - At Cathode (Reduction): $2H_2O+2e^- \longrightarrow 2OH^- + 2H^\bullet \longrightarrow H_2\uparrow$.
   - *Note:* Methane cannot be prepared by this method because the radical coupling step requires at least two $^\bullet CH_3$ radicals, giving a minimum of two carbons (ethane).

3.5 Physical Properties of Alkanes

- **Nature & Solubility:** Alkanes are non-polar molecules possessing weak intermolecular van der Waals forces. They are hydrophobic (insoluble in water) and soluble in non-polar organic solvents (*"like dissolves like"*).
- **Physical State at $298\,K$:** C_1 to C_4 are colourless, odourless gases; C_5 to C_{17} are liquids; C_{18} onwards are solids.
- **Boiling Point Trends:** Boiling point increases steadily with molecular mass due to increased surface area and stronger van der Waals forces. Among isomeric alkanes, **branching decreases the boiling point** because branched molecules become more spherical, reducing surface area of contact and weakening van der Waals forces.
- **Melting Point Trends:** Symmetrical isomers pack more tightly in the crystal lattice and have unusually high melting points (e.g., 2,2-dimethylpropane melts at $256.4\,K$ compared to $143.3\,K$ for *n*-pentane and $113.1\,K$ for isopentane).

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Molecular Formula	Alkane Name	Molecular Mass (u)	Boiling Point (K)	Melting Point (K)
CH_4	Methane	16	111.0	90.5
C_2H_6	Ethane	30	184.4	101.0
C_3H_8	Propane	44	230.9	85.3
C_4H_{10}	Butane (*n*-butane)	58	272.4	134.6
C_4H_{10}	2-Methylpropane	58	261.0	114.7
C_5H_{12}	Pentane (*n*-pentane)	72	309.1	143.3
C_5H_{12}	2-Methylbutane	72	300.9	113.1
C_5H_{12}	2,2-Dimethylpropane	72	282.5	256.4
C_6H_{14}	Hexane	86	341.9	178.5
C_7H_{16}	Heptane	100	371.4	182.4
C_8H_{18}	Octane	114	398.7	216.2
C_9H_{20}	Nonane	128	423.8	222.0
$C_{10}H_{22}$	Decane	142	447.1	243.3
$C_{20}H_{42}$	Eicosane	282	615.0	309.7

3.6 Chemical Properties of Alkanes

1. **Free Radical Substitution (Halogenation):** Takes place at high temperature (573–$773\ K$) or in diffused sunlight / UV light ($h\nu$).
   - Successive chlorination of methane yields $CH_3Cl \rightarrow CH_2Cl_2 \rightarrow CHCl_3 \rightarrow CCl_4$.
   - **Reactivity of Halogens:** $F_2 > Cl_2 > Br_2 > I_2$. Fluorination is too violent to control. Iodination is slow and reversible ($CH_4 + I_2 \rightleftharpoons CH_3I + HI$), so it must be carried out in the presence of oxidising agents like HIO_3 or HNO_3 which destroy HI ($HIO_3 + 5HI \rightarrow 3I_2 + 3H_2O$) and shift the equilibrium forward.
   - **Rate of Replacement of Hydrogen Atoms:** $3^\circ > 2^\circ > 1^\circ$ (due to stability order of alkyl free radicals).
   - **Free Radical Chain Mechanism:**
     - *Initiation:* Homolysis of $Cl-Cl$ bond (weaker than $C-C$ and $C-H$): $Cl_2 \xrightarrow{h\nu} 2Cl^\bullet$.
     - *Propagation:* $CH_4 + Cl^\bullet \xrightarrow{h\nu}{}^\bullet CH_3 + HCl$ followed by ${}^\bullet CH_3 + Cl_2 \xrightarrow{h\nu} CH_3Cl + Cl^\bullet$.
     - *Termination:* Combination of free radicals: $Cl^\bullet + Cl^\bullet \rightarrow Cl_2$; ${}^\bullet CH_3 + Cl^\bullet \rightarrow CH_3Cl$; and ${}^\bullet CH_3 + {}^\bullet CH_3 \rightarrow CH_3 - CH_3$. This last step explains why **ethane is formed as a minor byproduct during the chlorination of methane**.
2. **Combustion:** Complete oxidation in air/dioxygen gives CO_2, H_2O, and heat:

$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow n\,CO_2 + (n+1)\,H_2O$$

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l); \quad \Delta_c H^\circ = -890\,kJ\,mol^{-1}$$

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l); \quad \Delta_c H^\circ = -2875.84\,kJ\,mol^{-1}$$

   Incomplete Combustion: In insufficient air, alkanes form **carbon black** ($CH_4 + O_2 \rightarrow C(s) + 2H_2O$), used in ink, printer ink, black pigments, and filters.
3. **Controlled Catalytic Oxidation:**
   - $2CH_4 + O_2 \xrightarrow{Cu/523\,K/100\,atm} 2CH_3OH$ (Methanol)
   - $CH_4 + O_2 \xrightarrow{Mo_2O_3,\Delta} HCHO + H_2O$ (Methanal)
   - $2CH_3CH_3 + 3O_2 \xrightarrow{(CH_3COO)_2Mn,\Delta} 2CH_3COOH + 2H_2O$ (Ethanoic acid)
   - Alkanes ordinarily resist oxidation, but those having a **tertiary (3°) H atom** are oxidised by $KMnO_4$ to corresponding 3° alcohols:

$$(CH_3)_3CH \xrightarrow{KMnO_4\ (Oxidation)} (CH_3)_3COH \quad (2\text{-Methylpropan-2-ol})$$

4. **Isomerisation:** Heating *n*-alkanes with **anhydrous $AlCl_3$ and HCl gas** converts them into branched-chain isomers (e.g., *n*-hexane isomerises into 2-methylpentane and 3-methylpentane).
5. **Aromatization (Reforming):** Heating *n*-alkanes with six or more carbons at $773\,K$ under 10–$20\,atm$ pressure in the presence of oxides of vanadium (V_2O_5), molybdenum (Mo_2O_3), or chromium (Cr_2O_3) supported over alumina (Al_2O_3) causes dehydrogenation and cyclisation to benzene and its homologues (e.g., *n*-hexane → benzene + $4H_2$).
6. **Reaction with Steam:** At $1273\,K$ in the presence of Ni catalyst, methane reacts with steam to form carbon monoxide and dihydrogen (industrial preparation of H_2 gas):

$$CH_4 + H_2O \xrightarrow[1273\ K]{Ni} CO + 3H_2$$

7. **Pyrolysis (Cracking):** Thermal decomposition of higher alkanes into lower alkanes and alkenes on heating to $773\,K$ (or in the presence of $Pd/Pt/Ni$ catalysts). For example,hexane cracks at $773\,K$ to give $C_6H_{12} + H_2$, $C_4H_8 + C_2H_6$, and $C_3H_6 + C_2H_4 + CH_4$.

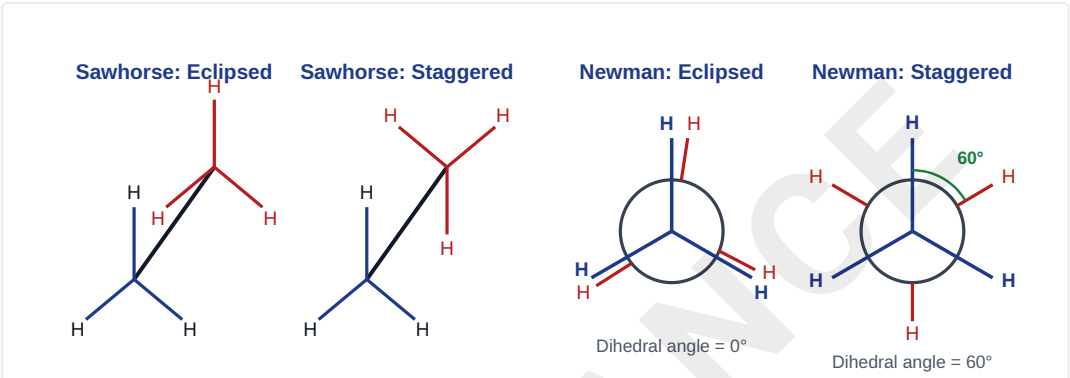
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Dodecane ($C_{12}H_{26}$, a constituent of kerosene oil) heated to $973\,K$ over $Pt/Pd/Ni$ gives heptane (C_7H_{16}), pentene (C_5H_{10}), and other products.

3.7 Conformations of Ethane

Because the electron distribution of a $C-C$ σ bond is cylindrically symmetrical around the internuclear axis, rotation around the $C-C$ single bond is permissible. Different spatial arrangements of atoms that can be converted into one another by rotation around a $C-C$ single bond are called **conformations**, **conformers**, or **rotamers**.

- **Torsional Strain:** Rotation is not completely free; it is hindered by a small energy barrier of $1–20\,kJ\,mol^{-1}$ ($12.5\,kJ\,mol^{-1}$ in ethane) due to weak repulsive interactions between adjacent bond electron clouds. This repulsion is called **torsional strain**.
- **Eclipsed Conformation:** Hydrogen atoms on the two carbons are as close together as possible (dihedral angle $\theta = 0^\circ$). Maximum torsional strain, highest energy, **least stable**.
- **Staggered Conformation:** Hydrogen atoms are as far apart as possible (dihedral angle $\theta = 60^\circ$). Minimum torsional strain, lowest energy, **most stable**.
- **Skew Conformations:** All infinite intermediate conformations between eclipsed and staggered.
- *Note:* Bond angles and bond lengths remain identical in all conformations. Because the $12.5\,kJ\,mol^{-1}$ barrier is easily overcome by thermal collisions at room temperature, conformers of ethane cannot be isolated separately.



Tutorial Diagram 2: Sawhorse and Newman projections of Eclipsed and Staggered conformations of ethane.

4. Alkenes (Olefins)

4.1 Structure of the Double Bond

Alkenes have the general formula C_nH_{2n} and are known as **olefins** (oil-forming) because ethene (C_2H_4) forms an oily liquid on reaction with chlorine. Methene (CH_2) has a very short life; ethene is the first stable member.

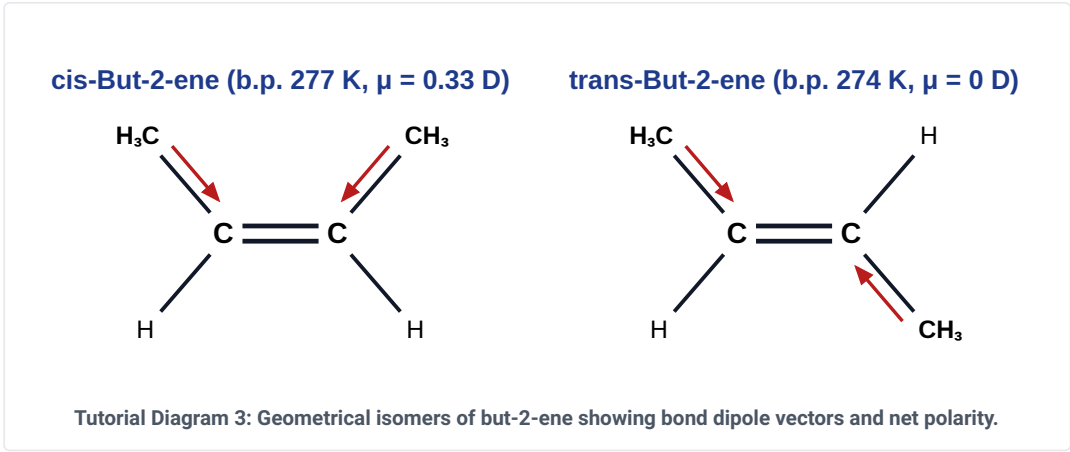
- **Bonding & Enthalpy:** The $C=C$ double bond (total bond enthalpy $\approx 681\,kJ\,mol^{-1}$, compared to $348\,kJ\,mol^{-1}$ for $C-C$ in ethane) consists of:
  - One strong **sigma (σ) bond** (bond enthalpy $\approx 397\,kJ\,mol^{-1}$) formed by head-on overlapping of sp^2 hybridised orbitals.
  - One weak **pi (π) bond** (bond enthalpy $\approx 284\,kJ\,mol^{-1}$) formed by lateral (sideways) overlapping of unhybridised $2p$ orbitals.
- **Geometry of Ethene:** Planar molecule; $C=C$ bond length = $134\,pm$, $C-H$ bond length = $110\,pm$, $H-C-H$ bond angle = 116.6°, and $H-C=C$ bond angle = 121.7°.
- **Reactivity:** Loosely held mobile π electrons make alkenes rich sources of electrons that are readily attacked by electron-deficient **electrophilic reagents**.

4.2 Structural & Geometrical Isomerism in Alkenes

- **Structural Isomerism in C_4H_8:**
  1. $^1CH_2=^2CH-^3CH_2-^4CH_3$ (But-1-ene)
  2. $^1CH_3-^2CH=^3CH-^4CH_3$ (But-2-ene)
  3. $^1CH_2=^2C(CH_3)-^3CH_3$ (2-Methylprop-1-ene)

  Here, (1) and (2) are **position isomers**, while (1) & (3) and (2) & (3) are **chain isomers**.
- **Geometrical (*cis*-*trans*) Isomerism:** Restricted rotation around the $C=C$ double bond gives rise to stereoisomers when each doubly bonded carbon is attached to two different atoms or groups ($YXC=CXY$, $XYC=CXZ$, or $XYC=CZW$).
  - ***cis*-Isomer:** Identical atoms/groups lie on the **same side** of the double bond. In *cis*-but-2-ene, the two $C-CH_3$ bond dipoles reinforce each other, giving a net dipole moment $\mu = 0.33\,D$ and a higher boiling point ($277\,K$) due to dipole-dipole interactions.
  - ***trans*-Isomer:** Identical atoms/groups lie on **opposite sides** of the double bond. In *trans*-but-2-ene, the two $C-CH_3$ bond dipoles cancel out ($\mu = 0$), giving a lower boiling point ($274\,K$) but a **higher melting point** due to better crystal packing in the solid state.

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Tutorial Diagram 3: Geometrical isomers of but-2-ene showing bond dipole vectors and net polarity.

4.3 Preparation of Alkenes

1. **Partial Reduction of Alkynes (Stereospecific):**
   - *Using Lindlar's Catalyst:* Reduction of an internal alkyne ($\mathrm{RC \equiv CR'}$) with $\mathrm{H_2}$ in the presence of palladised charcoal ($\mathrm{Pd/C}$) partially deactivated with poisons like sulphur compounds or quinoline gives a ***cis*-alkene** (*syn*-addition).
   - *Using Sodium in Liquid Ammonia (Birch Reduction):* Reduction of an internal alkyne with Na in liquid $\mathrm{NH_3}$ gives a ***trans*-alkene** (*anti*-addition).

2. **Dehydrohalogenation of Alkyl Halides (β-Elimination):** Heating an alkyl halide with **alcoholic** KOH eliminates HX ($-\mathrm{X}$ from α-carbon and $-\mathrm{H}$ from β-carbon):

$$\mathrm{R - CH_2 - CH_2 - X \xrightarrow[\Delta]{\text{alc. KOH}} R - CH = CH_2 + HX}$$

   - **Reactivity order of halogens:** $\mathrm{I > Br > Cl}$
   - **Reactivity order of alkyl groups:** $\mathrm{Tertiary\ (3^\circ) > Secondary\ (2^\circ) > Primary\ (1^\circ)}$

3. **Dehalogenation of Vicinal Dihalides:** Dihalides having two halogen atoms on adjacent carbon atoms (vicinal dihalides) react with Zn metal to lose $\mathrm{ZnX_2}$ and form an alkene:

$$\mathrm{CH_3 - CHBr - CH_2Br + Zn \longrightarrow CH_3 - CH = CH_2 + ZnBr_2}$$

4. **Acidic Dehydration of Alcohols (β-Elimination):** Heating alcohols with concentrated $\mathrm{H_2SO_4}$ eliminates a water molecule:

$$\mathrm{CH_3 - CH_2 - OH \xrightarrow[\Delta]{\text{Conc. } H_2SO_4} CH_2 = CH_2 + H_2O}$$

4.4 Physical & Chemical Properties of Alkenes

Physical Properties: $\mathrm{C_2}$ to $\mathrm{C_4}$ are gases, the next fourteen ($\mathrm{C_5}$–$\mathrm{C_{18}}$) are liquids, and higher ones are solids. Ethene has a faint sweet smell; all other alkenes are colourless and odourless. Every added $\mathrm{-CH_2-}$ group increases the boiling point by 20–30 K.

Chemical Reactions of Alkenes:

1. **Addition of Halogens (Test for Unsaturation):** $\mathrm{Br_2}$ or $\mathrm{Cl_2}$ adds across the double bond via a cyclic halonium ion intermediate to give vicinal dihalides. (Iodine does not react under normal conditions). The discharge of the **reddish-orange colour of bromine solution in** $\mathrm{CCl_4}$ serves as a laboratory test for unsaturation:

$$\mathrm{CH_2 = CH_2 + Br_2 \xrightarrow{CCl_4} BrCH_2 - CH_2Br \quad (1,2\text{-Dibromoethane})}$$

2. **Addition of Hydrogen Halides ($\mathrm{HI > HBr > HCl}$):**
   - **Markovnikov's Rule (Electrophilic Addition):** When an unsymmetrical reagent adds to an unsymmetrical alkene in the absence of peroxide, the **negative part of the addendum gets attached to the carbon atom possessing the lesser number of hydrogen atoms**.

     Mechanism: $\mathrm{H^+}$ attacks propene ($\mathrm{CH_3 - CH = CH_2}$) to form the more stable secondary carbocation ($\mathrm{CH_3 - \overset{+}{C}H - CH_3}$) rather than the less stable primary carbocation ($\mathrm{CH_3 - CH_2 - \overset{+}{C}H_2}$). Attack of $\mathrm{Br^-}$ on the 2° carbocation gives **2-bromopropane** as the major product.
   - **Anti-Markovnikov Addition / Peroxide Effect / Kharash Effect (Free Radical Mechanism):** In the presence of benzoyl peroxide $\mathrm{(C_6H_5CO)_2O_2}$, addition of HBr **only** to unsymmetrical alkenes occurs contrary to Markovnikov's rule, yielding **1-bromopropane** from propene.

     Mechanism:

     1. $\mathrm{(C_6H_5CO)_2O_2 \xrightarrow{Homolysis} 2C_6H_5COO^\bullet \longrightarrow 2^\bullet C_6H_5 + 2CO_2}$
     2. $\mathrm{^\bullet C_6H_5 + H - Br \xrightarrow{Homolysis} C_6H_6 + Br^\bullet}$
     3. $\mathrm{CH_3 - CH = CH_2 + Br^\bullet \longrightarrow CH_3 - \overset{\bullet}{C}H - CH_2Br}$ (more stable 2° free radical, major pathway) vs $\mathrm{CH_3 - CHBr - \overset{\bullet}{C}H_2}$ (less stable 1° radical).
     4. $\mathrm{CH_3 - \overset{\bullet}{C}H - CH_2Br + H - Br \longrightarrow CH_3 - CH_2 - CH_2Br + Br^\bullet}$ (1-Bromopropane, major product).

     Why only HBr*?* The $\mathrm{H - Cl}$ bond ($430.5\ \mathrm{kJ\ mol^{-1}}$) is stronger than the $\mathrm{H - Br}$ bond ($363.7\ \mathrm{kJ\ mol^{-1}}$) and cannot be cleaved by the phenyl radical. The $\mathrm{H - I}$ bond ($296.8\ \mathrm{kJ\ mol^{-1}}$) is weak, but the resulting iodine free radicals ($\mathrm{I^\bullet}$) rapidly combine to form $\mathrm{I_2}$ molecules instead of adding to the double bond.

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3. **Addition of Cold Concentrated H_2SO_4:** Follows Markovnikov's rule to form alkyl hydrogen sulphates ($CH_3 - CH = CH_2 + HOSO_2OH \rightarrow CH_3 - CH(OSO_2OH) - CH_3$).
4. **Acid-Catalysed Hydration:** Addition of H_2O in the presence of a few drops of conc. H_2SO_4 follows Markovnikov's rule (e.g., 2-methylpropene gives 2-methylpropan-2-ol).
5. **Oxidation of Alkenes:**
   - *Cold, dilute, aqueous $KMnO_4$ (Baeyer's Reagent at $273\,K$):* Oxidises alkenes to **vicinal glycols** (1,2-diols). Decolourisation of the purple $KMnO_4$ solution is a test for unsaturation:
   $$CH_2 = CH_2 + H_2O + (O) \xrightarrow[273\,K]{\text{dil. } KMnO_4} HO - CH_2 - CH_2 - OH \quad (\text{Ethane-1,2-diol})$$
   - *Acidic $KMnO_4$ or Acidic $K_2Cr_2O_7$:* Cleaves the double bond to give ketones and/or carboxylic acids (or CO_2 for terminal $= CH_2$):
   $$(CH_3)_2C = CH_2 \xrightarrow{KMnO_4/H^+} (CH_3)_2C = O + CO_2 + H_2O$$
   $$CH_3 - CH = CH - CH_3 \xrightarrow{KMnO_4/H^+} 2CH_3COOH$$
6. **Reductive Ozonolysis (O_3 followed by $Zn + H_2O$):** Ozone adds across the $C = C$ bond to form an **ozonide**, which is cleaved by Zn/H_2O into aldehydes and/or ketones. *Shortcut:* Replace the $C = C$ double bond with two $C = O$ double bonds ($> C = C < \xrightarrow{1.\ O_3\ \ 2.\ Zn/H_2O} > C = O + O = C <$). Used to locate the position of double bonds.
7. **Polymerisation:** At high temperature, high pressure, and in the presence of a catalyst, ethene polymerises to polythene $-(CH_2 - CH_2)_n-$ and propene polymerises to polypropene $-(CH(CH_3) - CH_2)_n-$.

5. Alkynes

5.1 Structure, Nomenclature, and Isomerism

Alkynes contain at least one carbon-carbon triple bond ($-C \equiv C-$) and have the general formula C_nH_{2n-2}. The first stable member is ethyne (C_2H_2), commonly called **acetylene** (used in oxy-acetylene flames for welding).

- **Structure of the Triple Bond in Ethyne:** Each carbon atom is sp **hybridised** (50% s-character). Ethyne is a **linear molecule** with an $H - C - C$ bond angle of 180°.
  - Contains **one** $C - C$ σ **bond** (sp–sp head-on overlap), **two** $C - H$ σ **bonds** (sp–$1s$ overlap), and **two mutually perpendicular** $C - C$ π **bonds** ($2p_x$–$2p_x$ and $2p_y$–$2p_y$ lateral overlap), forming a cylindrically symmetrical π-electron cloud around the internuclear axis.
  - **Bond Enthalpy:** $C \equiv C$ ($823\,kJ\,mol^{-1}$) > $C = C$ ($681\,kJ\,mol^{-1}$) > $C - C$ ($348\,kJ\,mol^{-1}$).
  - **Bond Length:** $C \equiv C$ ($120\,pm$) < $C = C$ ($133\,pm$) < $C - C$ ($154\,pm$).
- **Isomerism:** Ethyne and propyne have only one structure. Butyne (C_4H_6) has two position isomers: **but-1-yne** ($HC \equiv C - CH_2CH_3$) and **but-2-yne** ($CH_3 - C \equiv C - CH_3$).

5.2 Preparation of Alkynes

1. **From Calcium Carbide (Industrial Preparation of Ethyne):** Limestone is heated to quicklime, which is heated with coke to give calcium carbide (CaC_2). Hydrolysis of CaC_2 yields ethyne:
   $$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$
   $$CaO + 3C \longrightarrow CaC_2 + CO$$
   $$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$
2. **From Vicinal Dihalides (Double Dehydrohalogenation):** Treatment of 1,2-dibromoethane with **alcoholic** KOH removes one molecule of HBr to form a vinylic halide (bromoethene, $CH_2 = CHBr$), which requires the stronger base **sodamide** ($Na^+NH_2^-$) to eliminate the second HBr molecule and yield ethyne:
   $$BrCH_2 - CH_2Br \xrightarrow[-KBr,-H_2O]{\text{alc. } KOH} CH_2 = CHBr \xrightarrow[-NaBr,-NH_3]{NaNH_2} HC \equiv CH$$

5.3 Acidic Character & Chemical Properties of Alkynes

1. **Acidic Nature of Terminal Alkynes:** Hydrogen atoms attached directly to sp hybridised triply bonded carbon atoms ($-C \equiv C - H$) are acidic because sp hybridised carbon has the highest percentage of s-character (50%) and therefore the highest electronegativity, attracting the shared $C - H$ electron pair strongly and facilitating proton (H^+) release.
   - Terminal alkynes react with sodium metal or sodamide ($NaNH_2$) to form sodium acetylides/alkynides and liberate H_2 or NH_3:
   $$HC \equiv CH + Na \longrightarrow HC \equiv C^-Na^+ + \frac{1}{2}H_2 \uparrow \xrightarrow{+Na} Na^+C^- \equiv C^-Na^+ + \frac{1}{2}H_2 \uparrow$$
   $$CH_3 - C \equiv C - H + Na^+NH_2^- \longrightarrow CH_3 - C \equiv C^-Na^+ + NH_3$$
   - *Distinction Test:* Alkenes, alkanes, and non-terminal internal alkynes (like but-2-yne) do **not** show this reaction.
   - **Acidic Strength Order:**
     - $HC \equiv CH\ (sp, 50\%\,s) > H_2C = CH_2\ (sp^2, 33.3\%\,s) > CH_3 - CH_3\ (sp^3, 25\%\,s)$
     - $HC \equiv CH > CH_3 - C \equiv CH \gg CH_3 - C \equiv C - CH_3$
2. **Electrophilic Addition Reactions (via Vinylic Cation):**
   - *Addition of H_2:* $RC \equiv CH \xrightarrow{H_2,Pt/Pd/Ni} [RCH = CH_2] \xrightarrow{H_2} RCH_2CH_3$.

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- *Addition of Halogens (Br_2/CCl_4):* Decolourises reddish-orange Br_2 in CCl_4 to give a 1,1,2,2-tetrabromoalkane ($CH_3-C\equiv CH+2Br_2\rightarrow CH_3-CBr_2-CHBr_2$).
- *Addition of Hydrogen Halides (HX):* Two equivalents of HX add according to Markovnikov's rule to form **geminal dihalides** (both halogens on the same carbon):

$$HC\equiv CH\xrightarrow{HBr}[CH_2=CHBr]\xrightarrow{HBr}CH_3-CHBr_2\quad(1,1\text{-Dibromoethane})$$

$$CH_3-C\equiv CH\xrightarrow{HBr}[CH_3-CBr=CH_2]\xrightarrow{HBr}CH_3-CBr_2-CH_3\quad(2,2\text{-Dibromopropane})$$

- *Addition of Water (Kucherov Hydration):* Warming alkynes with **mercuric sulphate (Hg^{2+}) and dilute H_2SO_4 (H^+) at $333\,K$** adds one molecule of water to form an unstable enol that tautomerises (isomerises) into a carbonyl compound. Only **ethyne gives an aldehyde (ethanal)**; all other alkynes give ketones:

$$HC\equiv CH+H-OH\xrightarrow[333\,K]{Hg^{2+}/H^+}[CH_2=CH-OH]\xrightarrow{\text{Isomerisation}}CH_3-CHO\quad(\text{Ethanal})$$

$$CH_3-C\equiv CH+H-OH\xrightarrow[333\,K]{Hg^{2+}/H^+}[CH_3-C(OH)=CH_2]\xrightarrow{\text{Isomerisation}}CH_3-CO-CH_3\quad(\text{Propanone})$$

3. **Polymerisation of Ethyne:**
   - *Linear Polymerisation:* Produces polyacetylene $-(CH=CH-CH=CH)_n-$, a conducting polymer used as lightweight, inexpensive electrodes in batteries.
   - *Cyclic Polymerisation:* Passing ethyne through a **red-hot iron (Fe) tube at $873\,K$** trimerises three molecules of ethyne into **benzene** ($3C_2H_2\xrightarrow{\text{Red hot Fe tube, 873 K}}C_6H_6$).

6. Aromatic Hydrocarbons (Arenes)

6.1 Structure of Benzene, Resonance, and Aromaticity

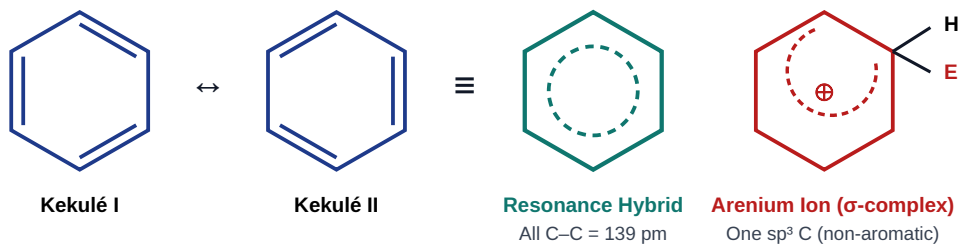
Benzene (C_6H_6) was isolated by Michael Faraday in 1825, and its cyclic structure with alternating single and double bonds was proposed by Friedrich August Kekulé in 1865.

- **Evidence for Kekulé Structure & Resonance:**
  - Benzene forms a **triozonide**, confirming three double bonds.
  - It yields only **one monosubstituted derivative** (all 6 carbons and 6 hydrogens are equivalent) and only **one *ortho*-disubstituted derivative** (1,2-dibromobenzene), explained by resonance between the two equivalent Kekulé structures.
  - **Orbital Picture & X-Ray Data:** All six carbon atoms are sp^2 hybridised in a planar hexagon. Each carbon has an unhybridised $2p$ orbital perpendicular to the ring plane, overlapping equally with both neighbours to form two continuous doughnut-shaped π-electron clouds above and below the ring plane. All six $C-C$ bond lengths are identical at $139\,pm$, intermediate between a $C-C$ single bond ($154\,pm$) and a $C=C$ double bond ($133\,pm$).

- **Conditions for Aromaticity (Hückel Rule):** A cyclic system (benzenoid or non-benzenoid) is aromatic if it satisfies three criteria:
  1. **Planarity** (all ring atoms sp^2 or sp hybridised).
  2. **Complete delocalisation** of π electrons in the ring.
  3. Presence of $(4n+2)$ π **electrons** in the ring, where n is a non-negative integer ($n=0,1,2,3,\ldots$, corresponding to $2,6,10,14,\ldots$ π electrons).

  Examples of Aromatic Species:
  - $n=1$ ($6\,\pi$ electrons): Benzene, **cyclopentadienyl anion** ($C_5H_5^-$), **cycloheptatrienyl cation** ($C_7H_7^+$, tropylium ion).
  - $n=2$ ($10\,\pi$ electrons): Naphthalene.
  - $n=3$ ($14\,\pi$ electrons): Anthracene, Phenanthrene.

- **Disubstituted Benzene Isomers:** 1,2- (or 1,6-) is *ortho* (*o*-); 1,3- (or 1,5-) is *meta* (*m*-); and 1,4- is *para* (*p*-). Dimethylbenzenes are called **xylenes** (*o*-, *m*-, and *p*-xylene).



Tutorial Diagram 4: Kekulé resonating structures, benzene resonance hybrid, and the arenium ion intermediate formed during electrophilic substitution.

6.2 Preparation of Benzene

1. **Cyclic Polymerisation of Ethyne:** $3HC\equiv CH\xrightarrow{\text{Red hot Fe tube, 873 K}}C_6H_6$.
2. **Decarboxylation of Aromatic Acids:** Heating sodium benzoate with soda lime ($NaOH+CaO$):

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$$C_6H_5COONa + NaOH \xrightarrow[\Delta]{CaO} C_6H_6 + Na_2CO_3$$

3. **Reduction of Phenol:** Passing phenol vapours over heated zinc dust:

$$C_6H_5OH + Zn \xrightarrow{\Delta} C_6H_6 + ZnO$$

6.3 Chemical Properties of Benzene

Aromatic hydrocarbons are non-polar, immiscible with water, and burn with a **sooty flame** due to high carbon content. General combustion equation for any hydrocarbon C_xH_y:

$$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \longrightarrow x\,CO_2 + \frac{y}{2}\,H_2O$$

A. Electrophilic Substitution Reactions (S_E)

Reaction	Reagents & Conditions	Active Electrophile (E^+)	Product
Nitration	Conc. HNO_3 + Conc. H_2SO_4, 323–333 K	Nitronium ion ($\overset{+}{N}O_2$) *(HNO_3 acts as a base towards H_2SO_4)*	Nitrobenzene ($C_6H_5NO_2$)
Halogenation	Cl_2 (or Br_2) + Anhyd. $AlCl_3$, $FeCl_3$, or $FeBr_3$	Chloronium ion (Cl^+)	Chlorobenzene (C_6H_5Cl) *(With excess $6Cl_2$/anhyd. $AlCl_3$ in dark, cold: Hexachlorobenzene C_6Cl_6)*
Sulphonation	Fuming sulphuric acid / Oleum $H_2SO_4(SO_3)$, Δ	SO_3 (or $^+SO_3H$)	Benzenesulphonic acid ($C_6H_5SO_3H$)
Friedel-Crafts Alkylation	$R-Cl$ + Anhyd. $AlCl_3$ (or $FeCl_3$)	Carbocation (R^+)	Alkylbenzene (C_6H_5R) *Note: $CH_3CH_2CH_2Cl$ gives isopropylbenzene (cumene) due to $1^\circ \rightarrow 2^\circ$ carbocation rearrangement!*
Friedel-Crafts Acylation	CH_3COCl or $(CH_3CO)_2O$ + Anhyd. $AlCl_3$, Δ	Acylium ion ($CH_3-\overset{+}{C}=O$)	Acetophenone ($C_6H_5COCH_3$)

Three-Step Mechanism of Electrophilic Substitution:

1. **Generation of Electrophile (E^+):**
   - $Cl_2 + AlCl_3 \rightarrow Cl^+ + [AlCl_4]^-$; $R-Cl + AlCl_3 \rightarrow R^+ + [AlCl_4]^-$; $CH_3COCl + AlCl_3 \rightarrow CH_3\overset{+}{C}=O + [AlCl_4]^-$.
   - In nitration: $HO_3SO-H + H-O-NO_2 \rightleftharpoons H-\overset{+}{O}(H)-NO_2 + HSO_4^- \rightleftharpoons H_2O + \overset{+}{N}O_2 + HSO_4^-$.
2. **Formation of Carbocation Intermediate (Arenium Ion / σ-Complex):** Attack of E^+ on the π ring forms a resonance-stabilised **arenium ion** in which the attacked carbon becomes sp^3 **hybridised**, interrupting cyclic delocalisation so the intermediate temporarily loses aromaticity.
3. **Removal of Proton (H^+):** Basic species ($[AlCl_4]^-$ or HSO_4^-) abstracts H^+ from the sp^3 carbon, restoring the aromatic ring and regenerating the catalyst.

B. Addition Reactions of Benzene (Vigorous Conditions)

- **Catalytic Hydrogenation:** $C_6H_6 + 3H_2 \xrightarrow{Ni, \Delta/\text{high pressure}} C_6H_{12}$ (Cyclohexane).
- **Free Radical Chlorination in UV Light:** Three molecules of Cl_2 add to benzene at 500 K under UV light to give **benzene hexachloride (BHC / gammaxane, $C_6H_6Cl_6$)**:

$$C_6H_6 + 3Cl_2 \xrightarrow[500\ K]{UV} C_6H_6Cl_6 \quad (1,2,3,4,5,6\text{-Hexachlorocyclohexane})$$

6.4 Directive Influence of Functional Groups in Monosubstituted Benzene

Category	Examples of Substituents	Electronic Effect & Ring Reactivity	Orientation of Incoming E^+
Activating Groups	$-OH, -NH_2, -NHR, -NHCOCH_3, -OCH_3, -CH_3, -C_2H_5$	$+R$ (or hyperconjugation/$+I$ for alkyl) outweighs $-I$. Increases overall electron density, especially at *ortho* and *para* positions.	***Ortho*- and *Para*- directing**
Moderately Deactivating Halogens	$-F, -Cl, -Br, -I$	Strong $-I$ effect decreases overall ring electron density (deactivating), but $+R$ resonance increases electron density at o- and p- relative to m-.	***Ortho*- and *Para*- directing**
Deactivating Groups	$-NO_2, -CN, -CHO, -COR, -COOH, -COOR, -SO_3H$	Strong $-I$ and $-R$ effects withdraw electrons strongly from o- and p- positions, leaving *meta* comparatively less electron-deficient.	***Meta*- directing**

6.5 Carcinogenicity and Toxicity

Benzene and polynuclear aromatic hydrocarbons (PAHs) containing more than two benzene rings fused together are toxic and **carcinogenic** (cancer-producing). They are formed during incomplete combustion of organic materials like tobacco, coal, and petroleum. In the human

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body, they undergo biochemical reactions that damage DNA and cause cancer. Key examples include: **1,2-benzanthracene**, **3-methylcholanthrene**, **1,2-benzpyrene**, **1,2,5,6-dibenzanthracene**, and **9,10-dimethyl-1,2-benzanthracene**.

7. Worked Examples (Chapter Master Problems)

Worked Example 1: Chain Isomers of C_6H_{14} and Structural Isomers of C_5H_{10} (Alkenes) & C_6H_{10} (Alkynes)

Problem: Write the IUPAC names and condensed structures of: (a) all 5 chain isomers of C_6H_{14}, (b) all 5 structural alkene isomers of C_5H_{10}, and (c) all 7 structural alkyne isomers of C_6H_{10}.

Solution:

- **(a) Five Chain Isomers of Hexane (C_6H_{14}):**
  1. $CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$: *n*-Hexane
  2. $CH_3-CH(CH_3)-CH_2-CH_2-CH_3$: 2-Methylpentane
  3. $CH_3-CH_2-CH(CH_3)-CH_2-CH_3$: 3-Methylpentane
  4. $CH_3-CH(CH_3)-CH(CH_3)-CH_3$: 2,3-Dimethylbutane
  5. $CH_3-C(CH_3)_2-CH_2-CH_3$: 2,2-Dimethylbutane

- **(b) Five Structural Alkene Isomers of C_5H_{10}:**
  1. $CH_2=CH-CH_2-CH_2-CH_3$: Pent-1-ene
  2. $CH_3-CH=CH-CH_2-CH_3$: Pent-2-ene
  3. $CH_3-C(CH_3)=CH-CH_3$: 2-Methylbut-2-ene
  4. $CH_3-CH(CH_3)-CH=CH_2$: 3-Methylbut-1-ene
  5. $CH_2=C(CH_3)-CH_2-CH_3$: 2-Methylbut-1-ene

- **(c) Seven Structural Alkyne Isomers of C_6H_{10}:**
  1. $HC\equiv C-CH_2-CH_2-CH_2-CH_3$: Hex-1-yne
  2. $CH_3-C\equiv C-CH_2-CH_2-CH_3$: Hex-2-yne
  3. $CH_3-CH_2-C\equiv C-CH_2-CH_3$: Hex-3-yne
  4. $HC\equiv C-CH(CH_3)-CH_2-CH_3$: 3-Methylpent-1-yne
  5. $HC\equiv C-CH_2-CH(CH_3)-CH_3$: 4-Methylpent-1-yne
  6. $CH_3-C\equiv C-CH(CH_3)-CH_3$: 4-Methylpent-2-yne
  7. $HC\equiv C-C(CH_3)_2-CH_3$: 3,3-Dimethylbut-1-yne

Worked Example 2: Correcting Incorrect IUPAC Names & Counting σ and π Bonds

Problem: (a) Give the correct IUPAC names for: (i) $(CH_3)_3CCH_2C(CH_3)_3$, (ii) tetra-*tert*-butylmethane, (iii) "2-Ethylpentane", and (iv) "5-Ethyl-3-methylheptane".
(b) Write the IUPAC names and calculate the number of σ and π bonds in:
(I) $(CH_3)_2CH-CH=CH-CH_2-CH=CH-CH(C_2H_5)CH_3$
(II) $CH_2=CH-CH=CH-CH=CH-CH=CH_2$
(III) $CH_2=C(CH_2CH_2CH_3)_2$
(IV) $CH_3-CH(CH_2CH_2CH_2CH_3)-CH=C(CH_2CH_3)-CH_2-CH(CH_3)_2$

Solution:

- **(a) Correct IUPAC Names:**
  - (i) Longest chain has 5 carbons with two methyls at $C2$ and two at $C4$: **2,2,4,4-Tetramethylpentane**.
  - (ii) Formula is $C[C(CH_3)_3]_4$. Longest chain is 5 carbons: **3,3-Di-*tert*-butyl-2,2,4,4-tetramethylpentane**.
  - (iii) In $CH_3-CH(C_2H_5)-CH_2-CH_2-CH_3$, the longest chain has 6 carbons, not 5. Correct name: **3-Methylhexane**.
  - (iv) In $CH_3-CH_2-CH(CH_3)-CH_2-CH(C_2H_5)-CH_2-CH_3$, numbering must start from the end giving the lower number to the alphabetically prior ethyl group. Correct name: **3-Ethyl-5-methylheptane**.

- **(b) IUPAC Names and σ/π Bond Counts:**
  - **(I)** Longest chain containing both double bonds is 10 carbons ($C_{12}H_{22}$): **2,8-Dimethyl-3,6-decadiene** (or 2,8-dimethyldeca-3,6-diene). Total σ bonds = $11\,(C\text{–}C)+22\,(C\text{–}H)=\mathbf{33}\,\sigma$ bonds; $\mathbf{2}\,\pi$ bonds.
  - **(II)** 8-carbon conjugated tetraene (C_8H_{10}): **1,3,5,7-Octatetraene** (octa-1,3,5,7-tetraene). Total σ bonds = $7\,(C\text{–}C)+10\,(C\text{–}H)=\mathbf{17}\,\sigma$ bonds; $\mathbf{4}\,\pi$ bonds.
  - **(III)** Longest chain containing the double bond is 5 carbons with a propyl group at $C2$ (C_8H_{16}): **2-*n*-Propylpent-1-ene**. Total σ bonds = $7\,(C\text{–}C)+16\,(C\text{–}H)=\mathbf{23}\,\sigma$ bonds; $\mathbf{1}\,\pi$ bond.
  - **(IV)** Longest chain with the double bond is 10 carbons ($C_{14}H_{28}$): **4-Ethyl-2,6-dimethyldec-4-ene**. Total σ bonds = $13\,(C\text{–}C)+28\,(C\text{–}H)=\mathbf{41}\,\sigma$ bonds; $\mathbf{1}\,\pi$ bond.

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Worked Example 3: Structure Determination via Ozonolysis

Problem: (a) An alkene 'A' on ozonolysis gives a mixture of ethanal and pentan-3-one. Write the structure and IUPAC name of 'A'.
(b) An alkene 'A' contains three $\mathrm{C-C}$, eight $\mathrm{C-H}$ σ bonds, and one $\mathrm{C-C}$ π bond. 'A' on ozonolysis gives two moles of an aldehyde of molar mass $44\ \mathrm{u}$. Write the IUPAC name of 'A'.
(c) Write down the ozonolysis products of 1,2-dimethylbenzene (*o*-xylene) and explain how this result supports Kekulé's oscillating structure for benzene.

Solution:

- **(a)** Align the carbonyl carbons of ethanal ($\mathrm{CH_3-CH=O}$) and pentan-3-one ($\mathrm{O=C(CH_2CH_3)_2}$) and replace the two $=\mathrm{O}$ atoms with a $\mathrm{C=C}$ double bond:
$$\mathrm{CH_3-CH=C(CH_2CH_3)_2} \implies \text{IUPAC Name: } \mathbf{3\text{-}Ethylpent\text{-}2\text{-}ene}$$
- **(b)** An aldehyde of molar mass $44\ \mathrm{u}$ is ethanal ($\mathrm{CH_3CHO}$, $12\times2+4\times1+16=44\ \mathrm{u}$). Joining two $\mathrm{CH_3CH=O}$ units gives $\mathrm{CH_3-CH=CH-CH_3}$ (**But-2-ene**), which indeed has three $\mathrm{C-C}$ σ bonds, eight $\mathrm{C-H}$ σ bonds, and one $\mathrm{C-C}$ π bond.
- **(c)** Ozonolysis of *o*-xylene gives a mixture of **three dicarbonyl products**:
  1. **Glyoxal** (Ethanedial, $\mathrm{OHC-CHO}$)
  2. **Methylglyoxal** (2-Oxopropanal, $\mathrm{CH_3-CO-CHO}$)
  3. **Dimethylglyoxal** (Butane-2,3-dione, $\mathrm{CH_3-CO-CO-CH_3}$)

  A single static Kekulé structure could only give two of these products; obtaining all three proves that both Kekulé structures contribute to the real benzene hybrid.

Worked Example 4: Multi-Step Conversions to Benzene & Disubstituted Benzenes

Problem: How will you convert: (a) Ethanoic acid into benzene, (b) Hexane into benzene, and (c) Benzene into (i) *p*-nitrobromobenzene, (ii) *m*-nitrochlorobenzene, (iii) *p*-nitrotoluene, and (iv) acetophenone?

Solution:

- **(a) Ethanoic acid to Benzene:**
$$\mathrm{CH_3COOH} \xrightarrow{\mathrm{NaOH}} \mathrm{CH_3COONa} \xrightarrow[\Delta]{\text{Soda lime}} \mathrm{CH_4} \xrightarrow{\mathrm{Cl_2/h\nu}} \mathrm{CH_3Cl} \xrightarrow[\text{Wurtz}]{\text{Na/dry ether}} \mathrm{C_2H_6} \xrightarrow{\mathrm{Cl_2/h\nu}} \mathrm{C_2H_5Cl}$$
$$\mathrm{C_2H_5Cl} \xrightarrow{\text{alc. KOH}} \mathrm{CH_2=CH_2} \xrightarrow{\mathrm{Br_2/CCl_4}} \mathrm{BrCH_2-CH_2Br} \xrightarrow{\text{alc. KOH}} \mathrm{CH_2=CHBr} \xrightarrow{\mathrm{NaNH_2}} \mathrm{HC\equiv CH} \xrightarrow[873\ \mathrm{K}]{\text{Red hot Fe tube}} \mathrm{C_6H_6}$$
- **(b) Hexane to Benzene (Aromatization):**
$$\mathrm{CH_3(CH_2)_4CH_3} \xrightarrow{\mathrm{Cr_2O_3\ or\ V_2O_5\ or\ Mo_2O_3,\ 773\ K,\ 10\text{–}20\ atm}} \mathrm{C_6H_6} + 4\mathrm{H_2}$$
- **(c) Conversions from Benzene (Order of Introducing Substituents):**
  - (i) ***p*-Nitrobromobenzene:** Since $-\mathrm{Br}$ is *ortho/para*-directing and $-\mathrm{NO_2}$ is *meta*-directing, brominate first, then nitrate:
$$\mathrm{C_6H_6} \xrightarrow{\mathrm{Br_2/Anhyd.\ FeBr_3}} \mathrm{C_6H_5Br} \xrightarrow{\mathrm{Conc.\ HNO_3+Conc.\ H_2SO_4,\ \Delta}} p\text{-}\mathrm{O_2N-C_6H_4-Br}\ \ (+\ o\text{-isomer, separated})$$
  - (ii) ***m*-Nitrochlorobenzene:** Nitrate first (since $-\mathrm{NO_2}$ is *meta*-directing), then chlorinate:
$$\mathrm{C_6H_6} \xrightarrow{\mathrm{Conc.\ HNO_3+Conc.\ H_2SO_4}} \mathrm{C_6H_5NO_2} \xrightarrow{\mathrm{Cl_2/Anhyd.\ AlCl_3}} m\text{-}\mathrm{O_2N-C_6H_4-Cl}$$
  - (iii) ***p*-Nitrotoluene:** Friedel-Crafts methylation first ($-\mathrm{CH_3}$ is *o/p*-directing), then nitration:
$$\mathrm{C_6H_6} \xrightarrow{\mathrm{CH_3Cl/Anhyd.\ AlCl_3}} \mathrm{C_6H_5CH_3} \xrightarrow{\mathrm{Conc.\ HNO_3+Conc.\ H_2SO_4}} p\text{-}\mathrm{O_2N-C_6H_4-CH_3}$$
  - (iv) **Acetophenone:** Friedel-Crafts acylation with acetyl chloride:
$$\mathrm{C_6H_6} + \mathrm{CH_3COCl} \xrightarrow{\mathrm{Anhyd.\ AlCl_3,\ \Delta}} \mathrm{C_6H_5COCH_3} + \mathrm{HCl}$$

8. NEET UG Exam-Style Practice Questions

Note: The following original practice questions are strictly aligned to the NEET UG examination standard to test conceptual clarity, reaction mechanisms, and problem-solving speed.

BOSONANCE

Practice Question 1 (Single-Correct MCQ — Nomenclature & Carbon Classification)

Consider the alkane $\mathrm{H_3C-CH_2-C(CH_3)_2-CH_2-CH(CH_3)_2}$. What are the total numbers of primary (1°), secondary (2°), tertiary (3°), and quaternary (4°) carbon atoms, and the number of hydrogen atoms bonded to 1°, 2°, and 3° carbons respectively?

(A) $1^\circ\ \mathrm{C}=5, 2^\circ\ \mathrm{C}=2, 3^\circ\ \mathrm{C}=1, 4^\circ\ \mathrm{C}=1$; Hydrogens: $1^\circ\ \mathrm{H}=15, 2^\circ\ \mathrm{H}=4, 3^\circ\ \mathrm{H}=1$

(B) $1^\circ\ \mathrm{C}=4, 2^\circ\ \mathrm{C}=3, 3^\circ\ \mathrm{C}=1, 4^\circ\ \mathrm{C}=1$; Hydrogens: $1^\circ\ \mathrm{H}=12, 2^\circ\ \mathrm{H}=6, 3^\circ\ \mathrm{H}=2$

(C) $1^\circ\ \mathrm{C}=5, 2^\circ\ \mathrm{C}=2, 3^\circ\ \mathrm{C}=2, 4^\circ\ \mathrm{C}=0$; Hydrogens: $1^\circ\ \mathrm{H}=15, 2^\circ\ \mathrm{H}=4, 3^\circ\ \mathrm{H}=2$

(D) $1^\circ\ \mathrm{C}=5, 2^\circ\ \mathrm{C}=1, 3^\circ\ \mathrm{C}=2, 4^\circ\ \mathrm{C}=1$; Hydrogens: $1^\circ\ \mathrm{H}=15, 2^\circ\ \mathrm{H}=2, 3^\circ\ \mathrm{H}=2$

Correct Answer: (A)

Worked Solution: In $\mathrm{H_3C-CH_2-C(CH_3)_2-CH_2-CH(CH_3)_2}$ (2,2,4-trimethylpentane + one $\mathrm{CH_2}$, i.e., 2,4,4-trimethylhexane):
• The five $-\mathrm{CH_3}$ terminal carbons are attached to only one carbon each → 5 **primary (1°) carbons** carrying $5\times3=15$ primary hydrogens.
• The two $-\mathrm{CH_2}-$ carbons are each bonded to two carbons → 2 **secondary (2°) carbons** carrying $2\times2=4$ secondary hydrogens.
• The $-\mathrm{CH}-$ carbon is bonded to three carbons → 1 **tertiary (3°) carbon** carrying 1 tertiary hydrogen.
• The central $-\mathrm{C}-$ atom is bonded to four carbons → 1 **quaternary (4°) carbon** carrying 0 hydrogens.

Practice Question 2 (Single-Correct MCQ — Conformations of Ethane)

Which of the following statements regarding the conformations of ethane is **incorrect**?

(A) The staggered conformation is more stable than the eclipsed conformation by about $12.5\ \mathrm{kJ\ mol^{-1}}$.

(B) The repulsive interaction between the electron clouds of adjacent $\mathrm{C-H}$ bonds that affects conformational stability is called torsional strain.

(C) Bond angles and bond lengths change significantly when ethane rotates from the staggered to the eclipsed conformation.

(D) It is not possible to separate and isolate different conformational isomers of ethane at room temperature.

Correct Answer: (C)

Worked Solution: In all conformations of ethane (eclipsed, staggered, and skew), the bond angles (109.5°) and bond lengths ($\mathrm{C-C}=154\ \mathrm{pm}$, $\mathrm{C-H}=112\ \mathrm{pm}$) remain exactly the same; only the dihedral (torsional) angle changes. Hence statement (C) is incorrect.

Practice Question 3 (Single-Correct MCQ — Physical Properties of Alkanes)

Arrange the three isomeric pentanes — pentane (I), 2-methylbutane (II), and 2,2-dimethylpropane (III) — in the correct order of their (i) decreasing boiling points and (ii) decreasing melting points:

(A) Boiling point: $\mathrm{I>II>III}$; Melting point: $\mathrm{I>II>III}$

(B) Boiling point: $\mathrm{I>II>III}$; Melting point: $\mathrm{III>I>II}$

(C) Boiling point: $\mathrm{III>II>I}$; Melting point: $\mathrm{III>I>II}$

(D) Boiling point: $\mathrm{I>III>II}$; Melting point: $\mathrm{I>III>II}$

Correct Answer: (B)

Worked Solution: With increasing branching, the molecule attains a spherical shape, reducing surface contact and intermolecular van der Waals forces. Thus boiling point decreases in the order: **pentane ($309.1\ \mathrm{K}$) > 2-methylbutane ($300.9\ \mathrm{K}$) > 2,2-dimethylpropane ($282.5\ \mathrm{K}$)** ($\mathrm{I>II>III}$). However, spherical symmetry allows tight packing in the solid crystal lattice, giving 2,2-dimethylpropane the highest melting point ($256.4\ \mathrm{K}$), followed by pentane ($143.3\ \mathrm{K}$) and 2-methylbutane ($113.1\ \mathrm{K}$) ($\mathrm{III>I>II}$).

BOSONANCE

Practice Question 4 (Assertion–Reason — Wurtz & Kolbe Methods)

Assertion (A): Neither Wurtz reaction nor Kolbe's electrolytic method is suitable for the preparation of methane.
Reason (R): Both reactions involve the coupling of two alkyl radicals (or two alkyl groups), so the product alkane must contain at least two carbon atoms.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: Both Wurtz reaction ($2\mathrm{R-X} + 2\mathrm{Na} \rightarrow \mathrm{R-R} + 2\mathrm{NaX}$) and Kolbe's electrolysis ($2\mathrm{RCOO^-} \rightarrow \mathrm{R-R} + 2\mathrm{CO_2} + 2\mathrm{e^-}$) form a new $\mathrm{C-C}$ bond between two alkyl units. Therefore, the simplest alkane that can be synthesised by either route is ethane ($\mathrm{C_2H_6}$), and methane ($\mathrm{CH_4}$) cannot be prepared. Both (A) and (R) are true and (R) correctly explains (A).

Practice Question 5 (Multiple-Statement — Controlled Oxidation of Alkanes)

Examine the following catalytic oxidation reactions of alkanes:

1. $2\mathrm{CH_4} + \mathrm{O_2} \xrightarrow{\mathrm{Cu/523\ K/100\ atm}} 2\mathrm{CH_3OH}$
2. $\mathrm{CH_4} + \mathrm{O_2} \xrightarrow{\mathrm{Mo_2O_3,\Delta}} \mathrm{HCHO} + \mathrm{H_2O}$
3. $2\mathrm{CH_3CH_3} + 3\mathrm{O_2} \xrightarrow{\mathrm{(CH_3COO)_2Mn,\Delta}} 2\mathrm{CH_3COOH} + 2\mathrm{H_2O}$
4. $\mathrm{(CH_3)_3CH} \xrightarrow{\mathrm{KMnO_4}} \mathrm{(CH_3)_3COH}$

Which of the above reactions are correctly matched with their products?

(A) 1 and 2 only

(B) 1, 2, and 3 only

(C) 2, 3, and 4 only

(D) 1, 2, 3, and 4

Correct Answer: (D)

Worked Solution: All four controlled oxidation reactions are standard reactions of alkanes: Cu at $523\ \mathrm{K}/100\ \mathrm{atm}$ oxidises methane to methanol; $\mathrm{Mo_2O_3}$ oxidises methane to methanal; manganese(II) acetate oxidises ethane to ethanoic acid; and $\mathrm{KMnO_4}$ oxidises tertiary $\mathrm{C-H}$ bonds (as in isobutane) to tertiary alcohols (2-methylpropan-2-ol).

BOSONANCE

Practice Question 6 (Single-Correct MCQ — Geometrical Isomerism)

Which of the following pairs of alkenes will exhibit *cis-trans* (geometrical) isomerism?

1. $(CH_3)_2C = CH - C_2H_5$
2. $CH_2 = CBr_2$
3. $C_6H_5CH = CH - CH_3$
4. $CH_3CH = CClCH_3$

(A) 1 and 2 only

(B) 3 and 4 only

(C) 1, 3, and 4 only

(D) 2 and 3 only

Correct Answer: (B)

Worked Solution: For an alkene to show geometrical isomerism, each doubly bonded carbon atom must be bonded to two *different* atoms or groups. In (1), one carbon has two identical $-CH_3$ groups. In (2), one carbon has two identical $-H$ atoms and the other has two identical $-Br$ atoms. In (3) $C_6H_5(H)C = C(H)CH_3$ and (4) $CH_3(H)C = C(Cl)CH_3$, each doubly bonded carbon carries two different groups, so 3 and 4 exhibit *cis-trans* isomerism.

Practice Question 7 (Single-Correct MCQ — Stereospecific Reduction & Markovnikov Addition)

But-2-yne is reduced with H_2 in the presence of Lindlar's catalyst to give compound **X**, whereas reduction of but-2-yne with Na in liquid NH_3 gives compound **Y**. Separately, hex-1-ene reacts with HBr in the absence of peroxide to give **P**, and in the presence of benzoyl peroxide to give **Q**. Identify **X, Y, P,** and **Q**:

(A) X = *cis*-but-2-ene, Y = *trans*-but-2-ene, P = 2-bromohexane, Q = 1-bromohexane

(B) X = *trans*-but-2-ene, Y = *cis*-but-2-ene, P = 1-bromohexane, Q = 2-bromohexane

(C) X = *cis*-but-2-ene, Y = *trans*-but-2-ene, P = 1-bromohexane, Q = 2-bromohexane

(D) X = butane, Y = *trans*-but-2-ene, P = 2-bromohexane, Q = 1-bromohexane

Correct Answer: (A)

Worked Solution: Lindlar's catalyst (Pd/C deactivated with sulphur/quinoline) performs *syn*-addition of H_2 to give ***cis*-but-2-ene (X)**. Sodium in liquid ammonia performs *anti*-addition to give ***trans*-but-2-ene (Y)**. Addition of HBr to hex-1-ene without peroxide follows Markovnikov's rule via a 2° carbocation to yield **2-bromohexane (P)**, while in the presence of peroxide it follows a free-radical anti-Markovnikov mechanism via a 2° free radical to yield **1-bromohexane (Q)**.

Practice Question 8 (Assertion–Reason — Kharash Peroxide Effect)

Assertion (A): The anti-Markovnikov addition of HX to unsymmetrical alkenes in the presence of benzoyl peroxide is observed only with HBr and not with HCl or HI.
Reason (R): The $H - Cl$ bond ($430.5\ kJ\ mol^{-1}$) is too strong to be cleaved homolytically by the free radical, whereas iodine free radicals formed from $H - I$ ($296.8\ kJ\ mol^{-1}$) combine with each other to form I_2 molecules instead of adding to the double bond.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: Only HBr (bond enthalpy $363.7\ kJ\ mol^{-1}$) has both propagation steps exothermic in the free-radical chain reaction. $H - Cl$ is too strong to be cleaved by the free radical, and $I^\bullet$ radicals from $H - I$ self-dimerise to I_2. Thus both (A) and (R) are true and (R) is the exact explanation of (A).

BOSONANCE

Practice Question 9 (Matching Type — Ozonolysis Products of Alkenes)

Match each alkene in **Column I** with its reductive ozonolysis products $(1.\ O_3;\ 2.\ Zn/H_2O)$ in **Column II**:

Column I (Alkene)	Column II (Ozonolysis Products)
(P) Pent-2-ene	(i) Butan-2-one + Pentan-2-one
(Q) 3,4-Dimethylhept-3-ene	(ii) Pentan-3-one + Methanal
(R) 2-Ethylbut-1-ene	(iii) Ethanal + Propanal
(S) 1-Phenylbut-1-ene	(iv) Benzaldehyde + Propanal

(A) P–(iii), Q–(i), R–(ii), S–(iv)

(B) P–(iii), Q–(ii), R–(i), S–(iv)

(C) P–(iv), Q–(i), R–(ii), S–(iii)

(D) P–(ii), Q–(i), R–(iii), S–(iv)

Correct Answer: (A)

Worked Solution: Cleaving the $C=C$ bond into two $C=O$ groups:
• **(P) Pent-2-ene** $(CH_3-CH=CH-CH_2CH_3)$ → CH_3CHO (ethanal) + CH_3CH_2CHO (propanal) [**iii**].
• **(Q) 3,4-Dimethylhept-3-ene** $(CH_3CH_2-C(CH_3)=C(CH_3)-CH_2CH_2CH_3)$ → $CH_3CH_2COCH_3$ (butan-2-one) + $CH_3COCH_2CH_2CH_3$ (pentan-2-one) [**i**].
• **(R) 2-Ethylbut-1-ene** $((CH_3CH_2)_2C=CH_2)$ → $(CH_3CH_2)_2C=O$ (pentan-3-one) + $HCHO$ (methanal) [**ii**].
• **(S) 1-Phenylbut-1-ene** $(C_6H_5-CH=CH-CH_2CH_3)$ → C_6H_5CHO (benzaldehyde) + CH_3CH_2CHO (propanal) [**iv**].

Practice Question 10 (Single-Correct MCQ — Hydrogenation & Structure Deduction)

How many different structural alkenes (excluding stereoisomers) on catalytic hydrogenation (H_2/Pd) yield **2-methylbutane**, and what is the structural formula of the alkene that gives **propanal and pentan-3-one** on reductive ozonolysis?

(A) 3 alkenes; $CH_3CH_2-CH=C(C_2H_5)_2$ (3-Ethylhex-3-ene)

(B) 4 alkenes; $CH_3-CH=C(C_2H_5)_2$ (3-Ethylpent-2-ene)

(C) 2 alkenes; $CH_3CH_2-CH=C(C_2H_5)_2$ (3-Ethylhex-3-ene)

(D) 3 alkenes; $CH_3CH_2-C(CH_3)=CH-CH_2CH_3$ (3-Methylhex-3-ene)

Correct Answer: (A)

Worked Solution:
1. The carbon skeleton of 2-methylbutane is $C^1-C^2(C)-C^3-C^4$. Placing a double bond in all possible distinct positions gives **3 alkenes**: (i) **2-methylbut-1-ene** $[CH_2=C(CH_3)-CH_2-CH_3]$, (ii) **2-methylbut-2-ene** [$CH_3-C(CH_3)=CH-CH_3]$, and (iii) **3-methylbut-1-ene** $[CH_3-CH(CH_3)-CH=CH_2]$.
2. Joining the carbonyl carbons of propanal $(CH_3CH_2CH=O)$ and pentan-3-one $(O=C(CH_2CH_3)_2)$ gives $CH_3CH_2-CH=C(CH_2CH_3)_2$, which is **3-ethylhex-3-ene**.

BOSONANCE

Practice Question 11 (Single-Correct MCQ — Acidic Behaviour of Hydrocarbons)

What is the correct decreasing order of acidic strength among **ethyne (I), benzene (II), and *n*-hexane (III)**, and what is the product formed when propyne is warmed with $\mathrm{Hg^{2+}/H^{+}}$ at $333\ \mathrm{K}$?

(A) $\mathrm{I > II > III}$; Propanal

(B) $\mathrm{I > II > III}$; Propanone

(C) $\mathrm{II > I > III}$; Propanone

(D) $\mathrm{I > III > II}$; Propan-2-ol

Correct Answer: (B)

Worked Solution: Electronegativity of carbon increases with percentage of s-character: Ethyne (sp, $50\%\ s$) > Benzene (sp^2, $33.3\%\ s$) > *n*-Hexane (sp^3, $25\%\ s$). Thus the decreasing order of acidity is **Ethyne (I) > Benzene (II) > *n*-Hexane (III)**. On hydration with $\mathrm{HgSO_4/dil.\ H_2SO_4}$ at $333\ \mathrm{K}$, propyne adds water according to Markovnikov's rule to give prop-1-en-2-ol, which tautomerises to **propanone** ($\mathrm{CH_3COCH_3}$).

Practice Question 12 (Single-Correct MCQ — Aromaticity & Hückel's Rule)

Consider the following cyclic species:
(I) Cyclopentadienyl anion ($\mathrm{C_5H_5^-}$)
(II) Cycloheptatrienyl cation ($\mathrm{C_7H_7^+}$)
(III) 6-Methylenecyclohexa-1,3-diene (a 6-membered ring with two endocyclic double bonds, one $-\mathrm{CH_2}-$ ring carbon, and one exocyclic $=\mathrm{CH_2}$ group)
(IV) Cyclopentadiene ($\mathrm{C_5H_6}$)
(V) Cyclooctatetraene ($\mathrm{C_8H_8}$)
Which of these species are **aromatic**?

(A) I, II, and III only

(B) I and II only

(C) I, II, and V only

(D) II, III, and IV only

Correct Answer: (B)

Worked Solution:
• **(I) Cyclopentadienyl anion ($\mathrm{C_5H_5^-}$)** and **(II) Cycloheptatrienyl cation ($\mathrm{C_7H_7^+}$)** are planar, fully conjugated cyclic systems containing $6\ \pi$ electrons ($4n+2$ with $n=1$), so both are **aromatic**.
• **(III)** and **(IV)** each contain an sp^3 hybridised $-\mathrm{CH_2}-$ ring carbon, which destroys planarity and prevents complete cyclic delocalisation of π electrons.
• **(V) Cyclooctatetraene ($\mathrm{C_8H_8}$)** has $8\ \pi$ electrons ($4n$, not $4n+2$) and adopts a non-planar tub conformation, so it is non-aromatic.

BOSONANCE

Practice Question 13 (Single-Correct MCQ — Electrophilic Substitution Reactivity)

Arrange the following sets of compounds in decreasing order of their relative reactivity towards an electrophile (E^+):
Set 1: Chlorobenzene (I), 2,4-Dinitrochlorobenzene (II), *p*-Nitrochlorobenzene (III)
Set 2: Toluene (P), *p*-$H_3C-C_6H_4-NO_2$ (Q), *p*-$O_2N-C_6H_4-NO_2$ (R)

(A) Set 1: $I > III > II$; Set 2: $P > Q > R$

(B) Set 1: $II > III > I$; Set 2: $P > Q > R$

(C) Set 1: $I > II > III$; Set 2: $Q > P > R$

(D) Set 1: $I > III > II$; Set 2: $R > Q > P$

Correct Answer: (A)

Worked Solution: The nitro group ($-NO_2$) is a strong electron-withdrawing ($-I, -R$) deactivating group, while the methyl group ($-CH_3$) is an electron-donating ($+I$, hyperconjugation) activating group.
• In **Set 1**, adding nitro groups progressively decreases ring electron density: **Chlorobenzene (I) > *p*-Nitrochlorobenzene (III) > 2,4-Dinitrochlorobenzene (II)**.
• In **Set 2**, replacing activating $-CH_3$ with deactivating $-NO_2$ decreases reactivity: **Toluene (P) > *p*-Nitrotoluene (Q) > *p*-Dinitrobenzene (R)**.

Practice Question 14 (Single-Correct MCQ — Friedel-Crafts Rearrangement & Chlorination Conditions)

What is the major organic product **A** formed when benzene is treated with 1-chloropropane in the presence of anhydrous $AlCl_3$, and what are the products **B** and **C** when benzene reacts with excess Cl_2 in the presence of (i) anhydrous $AlCl_3$ (dark, cold) and (ii) UV light at $500\,K$ respectively?

(A) A = *n*-Propylbenzene, B = C_6Cl_6, C = $C_6H_6Cl_6$

(B) A = Isopropylbenzene (cumene), B = C_6Cl_6 (hexachlorobenzene), C = $C_6H_6Cl_6$ (benzene hexachloride)

(C) A = Isopropylbenzene (cumene), B = $C_6H_6Cl_6$, C = C_6Cl_6

(D) A = *n*-Propylbenzene, B = C_6H_5Cl, C = C_6Cl_6

Correct Answer: (B)

Worked Solution:
• Reaction of 1-chloropropane ($CH_3CH_2CH_2Cl$) with anhydrous $AlCl_3$ initially generates a primary propyl carbocation ($CH_3CH_2\overset{+}{C}H_2$), which rapidly rearranges via a 1,2-hydride shift to the more stable secondary isopropyl carbocation ($CH_3\overset{+}{C}HCH_3$). Attack on benzene yields **isopropylbenzene (cumene, A)**.
• With excess Cl_2 and anhydrous $AlCl_3$ in the dark (electrophilic substitution), all six hydrogens are substituted to give **hexachlorobenzene (C_6Cl_6, B)**.
• With $3Cl_2$ under UV light at $500\,K$ (free radical addition), chlorine adds across the ring to give **benzene hexachloride ($C_6H_6Cl_6$, gammaxane, C)**.

9. Common Pitfalls & NEET Revision Checklist

Common Pitfalls to Avoid in NEET UG:

- **Pitfall 1 (Soda-lime Decarboxylation Stoichiometry):** Remember that soda lime decarboxylation removes the $-COONa$ group as Na_2CO_3, giving an alkane with **one carbon less**. To prepare propane (C_3H_8), you must start with *sodium butanoate* (C_3H_7COONa), not sodium propanoate!
- **Pitfall 2 (C_6Cl_6 vs $C_6H_6Cl_6$):** Do not confuse *hexachlorobenzene* (C_6Cl_6, formed by electrophilic substitution with excess Cl_2 /anhyd. $AlCl_3$ in the dark) with *benzene hexachloride* ($C_6H_6Cl_6$, formed by free-radical addition of $3Cl_2$ under UV light at $500\,K$).
- **Pitfall 3 (Halogens on Benzene Ring):** Halogens ($-F, -Cl, -Br, -I$) are **deactivating** due to their strong $-I$ effect, yet they are ***ortho/para*-directing** due to $+R$ resonance. Every other deactivating group ($-NO_2, -CN, -CHO, -COOH, -SO_3H$) is *meta*-directing.
- **Pitfall 4 (Alphabetising Alkyl Prefixes):** In IUPAC naming, hyphenated prefixes like *sec-* and *tert-* are ignored when alphabetising (so *sec*-butyl is alphabetised under 'b'), whereas non-hyphenated *iso* and *neo* are considered part of the group name (*isopropyl* is alphabetised under 'i').
- **Pitfall 5 (Nitration Acid-Base Role):** During generation of the nitronium ion ($\overset{+}{N}O_2$), concentrated H_2SO_4 acts as the **acid** (proton donor) and concentrated HNO_3 acts as the **base** (proton acceptor).

BOSONANCE

Concise Final Recap & Master Comparison Table

Feature	Alkanes	Alkenes	Alkynes	Arenes (Benzene)
General Formula	$\mathrm{C}_n\mathrm{H}_{2n+2}$	$\mathrm{C}_n\mathrm{H}_{2n}$	$\mathrm{C}_n\mathrm{H}_{2n-2}$	$\mathrm{C}_6\mathrm{H}_6$ ($4n + 2\,\pi\,\mathrm{e}^-$)
Hybridisation & Geometry	sp^3, Tetrahedral (109.5°)	sp^2, Trigonal planar ($\approx 120^\circ$)	sp, Linear (180°)	sp^2, Planar hexagon (120°)
C–C Bond Length & Enthalpy	$154\,\mathrm{pm}$ $348\,\mathrm{kJ\,mol^{-1}}$	$134\,\mathrm{pm}$ (or $133\,\mathrm{pm}$) $681\,\mathrm{kJ\,mol^{-1}}$	$120\,\mathrm{pm}$ $823\,\mathrm{kJ\,mol^{-1}}$	$139\,\mathrm{pm}$ (all equal) Resonance stabilised
Characteristic Reaction Type	Free radical substitution	Electrophilic addition	Electrophilic addition & terminal H acidity	Electrophilic substitution (S_E)
Bromine in $\mathrm{CCl_4}$ / Baeyer's Test	Negative (no decolourisation)	Positive (decolourises both)	Positive (decolourises $\mathrm{Br_2/CCl_4}$)	Negative under normal conditions
Reaction with Na or $\mathrm{NaNH_2}$	No reaction	No reaction	Terminal alkynes ($\mathrm{RC \equiv CH}$) liberate $\mathrm{H_2}$ / $\mathrm{NH_3}$	No reaction

BOSONANCE